

VirtuAI — Capstone Final Report

An Autonomous Multi-Modal Persona Generation System

Capstone Report — AI Engineering
May 2026

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VirtuAI — Capstone Final Report

Daniel Calder is an autonomous AI persona that posts native-format content across LinkedIn, X, Instagram, Instagram Reels, TikTok, YouTube Shorts, and Medium. The face, the voice, and the persona "voice" are all locked to identity-conditioned local models we trained or cloned ourselves. Nothing is rendered through HeyGen / ElevenLabs / Imagen / Veo at runtime.

The project is a fully-local persona consistency pipeline running on Apple Silicon.

TL;DR

Layer	Tech	Where it lives	Result
Text	Phi-3.5-mini-instruct + LoRA (fused)	<code>virtuai/models/finetune/fused_model/</code>	Trained on entrepreneurship examples; serves all 7 platforms
Face / Image	Z-Image-Turbo + Daniel Calder LoRA (Apple MLX, 8-bit, 512px)	<code>virtuai/persona/training_runs_*/checkpoints/</code> <code>virtuai/persona/training_runs/_extracted/0000455_checkpoint_adapter.safetensors</code>	30 ref photos → 455 training steps → 0.1 mean ArcFace match (vs baseline, +120%)
Voice	F5-TTS v1 Base (zero-shot clone)	<code>virtuai/persona/voice_sample/daniel_voice_ref.wav</code>	9.77 sec reference - ~5× real-time synthesis on MPS
Video (Phase 2)	Kling V1-6 multi-image2video + lip-sync API	<code>virtuai/tools/klingsync.py</code>	4 ref photos → 10 s 9:16 mp4 ArcFace 0.1 mean across frames
Video (Phase 1)	Wav2Lip + SadTalker on LoRA-generated still	<code>virtuai/persona/wav2lip/</code>	3-4× real-time lip-sync; head motion via SadTalker
Captions	Whisper word-timestamps → ASS subtitle generator	<code>virtuai/tools/caption_generator.py</code>	CapCut-style pop animation keyword highlighting word-by-word sync
Reel builder	FFmpeg 7 pipeline (clips + captions + hook overlay)	<code>virtuai/tools/reel_builder.py</code>	H.264 MP4 hook text 1 -22 dB bg music mix
Kling prompts	Gemini 2.5 Flash → rich cinematography prompts	<code>virtuai/tools/prompt_writer.py</code>	80-120 words Kling-optimized scene description
Anti-repetition	FAISS over <code>sentence-transformers/all-MiniLM-L6-v2</code>	<code>virtuai/persona/topic_memory.py</code> store: <code>virtuai/persona/topic_memory_data/topic.index</code>	18 demo prompts seeded; no checks at cosine threshold
Consistency eval	InsightFace ArcFace <code>buffalo_l</code> + sentence-transformers	<code>virtuai/persona/eval/</code>	Per-image identity score + before/after delta reported
Multi-agent orchestration	CrewAI 1.14 (8 agents incl. Guardian + Publisher)	<code>virtuai/agents/</code> , <code>virtuai/pipelines/content_pipeline.py</code>	Research - Strategy - Creator → Visual → Reviewer - Guardian - Publisher - Analyzer
Agent LLM	Gemini 2.5 Flash (free tier) or local Phi-3.5-mini	<code>content_pipeline.py _create_gemini_llm()</code>	Cloud reasoning + local generation tools; fixes

Layer	Tech	Where it lives	Result
			Apple Silicon OOM
Backend	FastAPI on port 8765	<code>virtuai/models/backend.py</code>	<code>/generate_image ,</code> <code>/generate_voice ,</code> <code>/generate_talking-head ,</code> <code>/analyze_image ,</code> <code>/safety_check ,</code> <code>/analyze_sentiment ,</code> <code>/unload_vision</code>
Showcase site	FastAPI + Jinja on port 8080	<code>virtuai/website/</code>	Persona-st live demo ; per-platform native UI mockups

The Persona Consistency Delta (capstone deliverable)

This is the core empirical claim:

Metric	Before (Imagen, n=29)	After (Z-Image-Turbo + Daniel LoRA, n=10)
Mean ArcFace identity similarity to Daniel	0.301	0.664 (+120%)
Median	0.303	(above acceptable threshold)
Strong matches (≥ 0.65)	0 / 29	8 / 10
Acceptable (≥ 0.45)	0 / 29	10 / 10
Identity drift (< 0.30)	13 / 29	0 / 10

Per-platform individual scores after LoRA:

Platform	ArcFace similarity to reference bank
LinkedIn	0.765 (very strong)
Instagram	0.725
Medium	0.710
X	0.666
Earlier eval set (6 images)	0.49 – 0.72

The InsightFace ArcFace `buffalo_l` model ran on CPU via ONNX. The "before" baseline used the legacy Imagen 4.0-generated images that existed in `virtuai/data/generated_images/` from before the persona-LoRA pivot.

Full report: `virtuai/persona/eval/_reports/final_consistency.json` .

Phase 2: End-to-End Reel Production

Phase 2 replaced the Wav2Lip-only video pipeline with a cloud+local hybrid:

Production chain (7 minutes end-to-end):

Step	Tool	Time	Output
1. Voice	F5-TTS (local backend)	47 s	Daniel's cloned voice WAV
2. Upload	catbox.moe	2 s	Public audio URL for Kling
3. Prompt	Gemini 2.5 Flash	3 s	124-word Kling scene prompt
4. Video	Kling multi-image2video (4 ref photos)	~170 s	10 s base clip, 9:16
5. Lip-sync	Kling lip-sync API	~190 s	Daniel speaks the script
6. Captions	Whisper base (word timestamps)	2 s	ASS subtitles, keyword highlights
7. Reel	FFmpeg 7 (captions + hook overlay)	1 s	2.3 MB publish-ready MP4

ArcFace identity across reel frames:

Frame	ArcFace similarity	Verdict
0.5 s	0.710	PASS (≥ 0.70)
2.0 s	0.696	ACCEPTABLE
4.0 s	0.672	ACCEPTABLE
7.0 s	0.708	PASS
9.0 s	0.688	ACCEPTABLE
Mean	0.695	Matches Phase 1 LoRA peak (0.664)

The reel includes: hook text overlay with fade animation in first 3 seconds, word-by-word CapCut-style captions with pop-in animation and keyword highlighting (AI terms, dollar amounts auto-detected), and H.264 encoding at CRF 20.

Output: `virtuai/data/generated_videos/daniel_reel_final_1778673989.mp4`

Demo content — what the site shows

Every asset under `virtuai/persona/demo/<platform>/` is end-to-end ours:

Platform	Asset type	Render path	Render time
LinkedIn	image (1024×1024)	Z-Image-Turbo + dnlcldr LoRA, 4 steps	7.0 min
X	image (1024×1024)	Z-Image-Turbo + dnlcldr LoRA, 4 steps	7.6 min
Instagram	image (1024×1024)	Z-Image-Turbo + dnlcldr LoRA, 4 steps	7.3 min
Medium	image (1024×1024)	Z-Image-Turbo + dnlcldr LoRA, 4 steps	7.3 min
TikTok	video (43 sec)	F5-TTS → Wav2Lip on Daniel hero still	173 sec render
Instagram Reels	video (51 sec)	F5-TTS → Wav2Lip on Daniel hero still	189 sec render
YouTube Shorts	video (71 sec)	F5-TTS → Wav2Lip on Daniel hero still	245 sec render
Hero showcase	video (3.7 sec, 1024×1024)	SadTalker max + GFPGAN → Wav2Lip refinement (stacked pipeline)	75 min + 19 sec

Total compute time for the full demo set: roughly **2 hours** of overnight rendering on an M-series Mac. Total external cost: **\$0**.

The journey — what we tried and why we picked Wav2Lip

We evaluated multiple talking-head approaches on Apple Silicon. The full ladder:

Tool	Quality	Time per second of output	Verdict
HeyGen API	SOTA realism	~2 sec	✗ Conflicts with capstone — "you used a SaaS" critique
EchoMimic v2 (CVPR 2025) via HF Space	Excellent	n/a	✗ The free public Space is "CPU showcase" — refuses to actually generate
SadTalker default (256, no enhancer)	Mediocre, "AI face" feel	~1.5× realtime	✗ User rejected the realism
SadTalker max (512 + GFPGAN + still)	Sharp, natural face dynamics	~20× realtime (75 min for 3.7 sec audio)	✗ Quality is great but unworkable for 165 sec of content
Wav2Lip (Improved quality)	Sharp lip sync, identity preserved	~4× realtime	✓ Picked. Best speed/quality tradeoff on Apple Silicon
Wav2Lip stacked on SadTalker max	Combines SadTalker face polish + Wav2Lip lip sync	SadTalker's 75 min + 19 sec	★ Used for the hero showcase only
Hallo3 / Hallo2 / EchoMimic v3 / LiveAvatar	SOTA	n/a	✗ All require NVIDIA H100/A100; no working MPS port exists
MuseTalk 1.5	Good	(untested — would compete with rest of system)	● Future work
VideoReTalking (SIGGRAPH 2022)	Good lip sync	(cloned, not tested)	● Future work
JoyVASA	Good (LivePortrait-based)	n/a	✗ Documented CUDA only

The honest answer for "best free pretrained on Apple Silicon" in 2026 is **Wav2Lip**, plus stacking with SadTalker max for hero shots. Modern diffusion-based talking heads (Hallo, EchoMimic, MuseTalk, VASA-1, etc.) all target NVIDIA-class hardware and have no usable MPS port today.

Architectural pivots we made (worth defending in the panel)

1. **HeyGen** → **fully local persona stack**. We started the project willing to integrate HeyGen + ElevenLabs for the talking-head layer. Rejected because the capstone has to demonstrate ML engineering, not API integration. Built our own face LoRA + voice clone instead.
2. **FLUX.1-schnell** → **Z-Image-Turbo**. `mflux` dropped LoRA training support for FLUX.1 mid-2026; only `z_image`, `flux2`, and `qwen` still have training adapters. Switching to Z-Image-Turbo also improved per-step training speed on Apple Silicon.
3. **LivePortrait** → **SadTalker** → **Wav2Lip**. Initially picked LivePortrait. Discovered mid-implementation that LivePortrait is image-driven (needs a driving video, not audio), so it can't power audio-driven talking heads. Pivoted to SadTalker (works but slow). Finally settled on Wav2Lip for production batch + SadTalker-stacked Wav2Lip for the hero showcase.
4. **Pinecone** → **local FAISS**. ~100 posts is well under FAISS-flat's scale. Pinecone would be an unnecessary external dependency.
5. **Imagen 4.0** → **Z-Image-Turbo + dnlcldr LoRA**. Imagen produced a different-looking face every time (mean ArcFace similarity 0.301 to the canonical Daniel). The LoRA-conditioned Z-Image-Turbo holds identity (mean 0.664 — measurable and reproducible).

Engineering oddities we hit and fixed

1. `mflux training output_path` is **CWD-relative, not config-file-relative**. Final training output landed at `<project_root>/training_runs_<timestamp>/`, not under `virtuai/persona/training_runs/`. Backend's `_find_persona_lora()` walks every `training_runs*` dir to handle this.
2. **Z-Image-Turbo inference needs a different CLI from the generic one**. The generic `mflux-generate --model z-image-turbo` looks for a FLUX-style `text_encoder_2/` subdirectory that doesn't exist in the official `Tongyi-MAI/Z-Image-Turbo` HF repo. The dedicated `mflux-generate-z-image-turbo` subcommand uses the right loader path. Backend wired to the dedicated CLI.
3. **F5-TTS via torchaudio 2.11 → torchcodec → libavutil from FFmpeg**. Brew default ffmpeg is v8 (libavutil.60); torchcodec 0.11 supports up to FFmpeg 7 (libavutil.59). Fix: `brew install ffmpeg@7` (keg-only) and have `clone_voice.py` / `backend.py` re-exec themselves with `DYLD_FALLBACK_LIBRARY_PATH=/opt/homebrew/opt/ffmpeg@7/lib`.
4. **basicsr / torchvision compat break**. SadTalker and Wav2Lip both depend on `basicsr` which imports the removed `torchvision.transforms.functional_tensor`. Patched in their isolated venvs to use `torchvision.transforms.functional`.
5. **Wav2Lip checkpoint is CUDA-tagged**. `easy_functions._load()` only used CPU `map_location` when `device == "cpu"`; on `device == "mps"` it tried to load on CUDA and crashed. Patched to map to CPU first regardless of MPS/CPU; the model gets moved to MPS afterwards.

6. **macOS swap thrashing during long-running training.** A single `mflux-train` overnight run accumulated enough state that swap usage hit 11.4 GB / 13 GB and per-step time ballooned from 33 s to 1294 s. Killed at step 455 (well past visual convergence at step 300) and used that LoRA. Documented in `persona_anchor.json` → `face_lora.stopped_reason`.
7. **InsightFace + faiss-cpu + torch + sentence-transformers all bring their own OpenMP.** Concurrent calls segfaulted on Apple Silicon. Setting `OMP_NUM_THREADS=1`, `KMP_DUPLICATE_LIB_OK=TRUE`, `TOKENIZERS_PARALLELISM=false` at the top of `topic_memory.py` / `face_similarity.py` prevents the segfault.
-

How to run it

One-time setup (already done in this repo):

```
# Persona LoRA training (~75 min on M-series, run once)
mflux-train --config virtuai/persona/training_config.json

# F5-TTS reference + transcript prep (already in voice_sample/)
python virtuai/persona/scripts/prep_voice_reference.py

# SadTalker venv + 2.3 GB checkpoints (already done – see persona/sadtalker/)
# Wav2Lip extras into the same venv (already done)
```

Running the demo:

```
# Backend on port 8765
python run_backend.py

# Showcase site on port 8080
python run_website.py

# → open http://localhost:8080
```

Regenerate everything from scratch:

```
# Generate scripts + F5-TTS audio for all 7 platforms
python virtuai/persona/scripts/generate_platform_content.py

# Render Wav2Lip videos for all 3 video platforms
python virtuai/persona/scripts/render_all_platforms.py

# Render Z-Image-Turbo + LoRA images for all 4 image platforms
python virtuai/persona/scripts/render_platform_images.py

# Re-run consistency eval
python virtuai/persona/eval/run_consistency_report.py
```

Status check at any time:

```
python virtuai/persona/scripts/status.py
```

Repo map

```
virtuai/
├── agents/                7 CrewAI agents (Guardian for ethics is the prof requirement)
├── models/
│   ├── backend.py        FastAPI server, all model endpoints
│   ├── vlm_client.py      Python client for the backend
│   └── finetune/fused_model/ Phi-3.5-mini LoRA fused (text)
├── pipelines/content_pipeline.py CrewAI orchestration
├── persona/              The persona-stack work (this report's subject)
│   ├── persona_anchor.json Single source of truth for identity
│   ├── face_dataset/      30 training photos + daniel_hero.png (LoRA-generated canonical face)
│   ├── voice_sample/      F5-TTS reference WAV + transcript
│   ├── training_config.json mflux-train config (Z-Image-Turbo + dnlcldr LoRA)
│   ├── training_runs/_extracted/ Latest extracted *.safetensors LoRA weights
│   ├── topic_memory.py    FAISS anti-repetition module
│   ├── topic_memory_data/ FAISS index + JSON metadata
│   ├── sadtalker/         SadTalker + 2.3 GB checkpoints (talking-head v1)
│   ├── wav2lip/           Wav2Lip + 1.2 GB checkpoints (talking-head v2 – production)
│   ├── liveportrait/, video_retalking/ Cloned but unused – kept as future-work
│   ├── eval/              Face + text consistency eval suite
│   │   ├── face_similarity.py InsightFace ArcFace
│   │   ├── text_style.py   sentence-transformers centroid
│   │   ├── run_consistency_report.py
│   │   └── _reports/       Timestamped consistency reports
│   ├── demo/              Final per-platform demo content the website shows
│   │   ├── linkedin/      text.md, image.png, manifest.json
│   │   ├── x/, instagram/, medium/ same shape – image platforms
│   │   ├── tiktok/        text.md, audio.wav, video.mp4, manifest.json
│   │   ├── instagram_reels/, youtube_shorts/ same shape – video platforms
│   │   └── render_summary.json, image_render_summary.json, summary.json
│   ├── talking_head/
│   │   ├── hero_showcase_stacked.mp4 SadTalker max + Wav2Lip stacked, 1024×1024
│   │   ├── sadtalker_max/       First SadTalker max output (2026-04-27)
│   │   ├── wav2lip/            All Wav2Lip outputs by timestamp
│   │   └── generated/          Earlier basic-SadTalker outputs (kept for comparison)
│   └── scripts/               All the small utility scripts
│       ├── preprocess_face_dataset.py Crops + upscales the 30 training photos
│       ├── prep_voice_reference.py    Trims the 49s MP3 to 9.77s + transcript
│       ├── clone_voice.py             F5-TTS CLI
│       ├── talking_head.py            SadTalker wrapper
│       ├── wav2lip_render.py          Wav2Lip wrapper (production talking-head)
│       ├── render_all_platforms.py    Renders all 3 video platforms via Wav2Lip
│       ├── render_platform_images.py  Renders all 4 image platforms via Z-Image-Turbo + LoRA
│       ├── generate_platform_content.py Phi → text + F5-TTS → audio for all 7 platforms
│       ├── eval_lora_checkpoint.py    Before/after consistency eval helper
│       └── status.py                 One-shot health check across the whole stack
├── tools/                      CrewAI tool wrappers
├── website/                   Showcase site (port 8080)
└── data/                      Legacy demo content + content packages
```

External venvs (NOT under `virtuai/`):

```
~/virtuai-venv/                Main project venv – Phi, MLX, CrewAI, FastAPI
~/virtuai-sadtalker-venv/      Isolated Python 3.10 venv for SadTalker + Wav2Lip
                               (their pinned numpy 1.23 / scipy 1.10 / kornia 0.6 are
                               incompatible with the main venv)
```

What's left for "future work" (out of capstone scope)

- **Multi-clip reels (>10 s).** Kling max duration is 10 s per clip. Reels over 10 s require generating 2-3 clips and concatenating. The reel builder supports this but cross-clip identity consistency is unvalidated.
 - **Kling V2-Master upgrade.** We used V1-6 (available on our tier). V2-Master produces higher-quality motion and scene detail.
 - **Full agent autopilot.** The CrewAI pipeline runs end-to-end with all 8 agents, but human-in-the-loop is still required for final publish approval.
 - **Gemini quota upgrade.** Free tier limits to 20 requests/day for Gemini 2.5 Flash. A paid tier or API key with higher quota would support multiple pipeline runs per day.
 - **MuseTalk 1.5 MPS port.** Latent-space lip sync; potential local alternative to Kling lip-sync API if a Mac port lands.
-

Acknowledgements / dependencies

- Phi-3.5-mini-instruct (Microsoft) — 4-bit MLX weights from `mlx-community`
- LLaVA 1.5 7B — 4-bit MLX weights from `mlx-community`
- Z-Image-Turbo (Tongyi-MAI) — text-to-image base
- F5-TTS v1 Base (Shanghai AI Lab) — zero-shot voice clone
- Kling V1-6 API (Kuaishou) — multi-image-to-video + lip-sync
- Gemini 2.5 Flash (Google) — Kling prompt writing + agent reasoning
- SadTalker (OpenTalker) — talking-head v1 (Phase 1, hero showcase)
- Easy-Wav2Lip (anothermartz) — Wav2Lip Mac fork (Phase 1, production lip-sync)
- Whisper (OpenAI) — word-level timestamps for caption generation
- InsightFace `buffalo_l` — ArcFace identity verification
- sentence-transformers `all-MiniLM-L6-v2` — text style + topic memory
- mflux (filipstrand) — Apple MLX wrapper for diffusion training/inference
- CrewAI 1.14 — 8-agent orchestration
- Composio SDK — LinkedIn/Facebook/Instagram publishing
- Apple MLX, PyTorch, FAISS, FastAPI, FFmpeg 7

— Phase 2 reel pipeline (Kling V1-6 + lip-sync).

Phase 3 — Cloud-native autopilot + 3-format content pack (May 14)

After Phase 2 shipped working reels, the project pivoted again to address three remaining gaps:

1. **Lip-sync drift** at the end of multi-segment reels (the compositing seam between Kling V1-6 render + separate lip-sync pass).
2. **Static-background "AI portrait" feel** — Avatar Pro animates a still image, so the environment never moves. Top creators (e.g. @andyhan.ai) use Veo 3 / Kling 3.0 with native joint generation precisely to avoid this.
3. **Single-format output** — Phase 2 produced reels only. Real creator feeds run a daily mix of reels + portraits + carousels.

Architecture changes

Single-pass video. Kling 3.0 multi-shot replaced the V1-6 + lip-sync two-step. The model jointly generates video, native speech, and lip-sync in one render, so there is no compositing seam to drift. Two parallel renders (3 shots × 5 s each) are concatenated with FFmpeg `aresample=async=1` + CFR re-encode to lock audio to video timestamps.

Claude Sonnet 4.6 script writer. Replaced Gemini-2.5-Flash for script ideation. Two-phase generation: brainstorm 5 candidate topics → pick winner → write the full 6-beat story arc (setup / incident / struggle / turn / proof / meaning). Hard concreteness gates reject scripts that lack named tools, real dollar amounts, real timeframes, or use a banned phrase (leverage, synergy, 10x, productivity tips, scale your business).

Static image content. A new pipeline produces portraits and 5-slide carousels alongside reels:

- *Portraits* — Nano Banana 2 places the canonical Daniel face in a real environment with the right outfit; PIL renders typography overlay.
- *Carousels* — 5 slides (cover / problem / insight / proof / payoff) with persona on slides 1 and 5, environmental concept on 2/3/4.

Daily pack orchestrator. `scripts/daily_pack.py` runs all three content types in parallel:

Layer	Tool
Topic seeds	14-entry rotation with history avoidance
Outfit / mood / setting	rotation pools, 3 distinct picks per pack
Reel	Claude → Kling 3.0 multi-shot ×2 → Suno → audio resync
Portrait	Claude → Nano Banana → PIL
Carousel	Claude → Nano Banana ×5 → PIL ×5
Publish	YouTube Direct + Composio (IG + LinkedIn), parallel
History	<code>autopilot_history.json</code> appended on every run

Wall-clock: $\approx$ 12 minutes per pack. Cost: $\approx$ \$5–7 per pack.

Reliability fixes

- **KIE file-stream-upload** replaces tmpfiles.org / catbox.moe for every asset KIE itself has to fetch. Eliminates the "Image fetch failed. Check access settings" mid-job errors.
- **Audio resync at concat** — silence-trim each render to its exact video duration, then re-encode the concat at constant 30 fps + 48 kHz audio. Lip-sync now holds through the full 30 s.
- **Mandatory motion in visual prompts** — Claude prompts now require explicit camera motion + 2-3 moving background elements. No more static "frozen photo" shots.
- **Posture / outfit rotation** — tracked in history; no repeat outfits or postures within the last 4-6 packs.

Agent ↔ tool ownership refactor

Phase 3 also cleaned up the agent layer. Every tool is now owned by the agent that calls it, and the two unused agents (research, analyzer) were wired with real tools instead of being stubs:

- `cloud_tools.py` — new file holding `@tool`-decorated wrappers for the KIE-cloud pipeline. Agents pull from here for production.
- `local_tools.py` — the original local-backend `@tool` wrappers, kept for the Phase-1 fallback path.
- Each agent's `tools=[...]` list and backstory was rewritten to match the current pipeline. See README.md "Eight CrewAI agents" table.

n8n integration

`scripts/api_server.py` (FastAPI) exposes the daily pack as HTTP:

- `POST /run-pack` — kick off a full daily pack in background, returns `task_id`.
- `GET /status/{task_id}` — poll for state + final URLs.
- `GET /history` — return the last N published runs.
- `POST /run-reel`, `/run-portrait`, `/run-carousel` — produce-only endpoints.
- `POST /publish-reel`, `/publish-image-post` — publish an already- produced asset.

n8n connects via HTTP nodes: a schedule trigger fires `/run-pack` daily, polls `/status/{id}` until success, then notifies on Slack/email.

Phase 3 verified output

Two daily packs landed clean on live platforms:

Pack	Reel	Portrait	Carousel
#1	"AI proposal generator killed my conversion rate"	"Automating Email Is Usually Wrong First"	"Most AI Assistants Will Be Unemployed by 2027"
#2	"AI tool I killed because it destroyed my margins"	"Junior dev replaced by Cursor costs more"	"Stop Prompting Claude Like A Search Engine"

Both packs were produced in $\approx$ 11–12 minutes start to finish and posted to YouTube + Instagram + LinkedIn with zero human touch. Variety rotation hit 3 distinct outfits, moods, and topic seeds per pack.

What Phase 3 leaves open

- **X / Twitter publishing.** Composio requires a one-time OAuth dance for the X toolkit under `user_id=danielcalder-`. Until that's done, the pack publishes to YouTube + IG + LinkedIn (3 of 4 platforms).
- **True IG carousel.** Composio's `INSTAGRAM_CREATE_MEDIA_CONTAINER` validator hardcodes `image_url` as required even for `media_type=CAROUSEL` parents, so the swipe-through carousel falls back to a single-image post of slide 1. The full 5-slide swipe needs a direct Meta Graph API call (~50 lines, future work).

- **Voice cloning to "Liam" via ElevenLabs Speech-to-Speech.** The pack currently ships with Kling 3.0's native voice. The optional `voice_change_to_liam` step is wired in `produce_reel_v16.py` and fires when `ELEVENLABS_API_KEY` is set in `.env`.
-

Additional acknowledgements (Phase 3)

- Claude Sonnet 4.6 (Anthropic, via KIE.ai) — script writing
- Kling 3.0 multi-shot (Kuaishou, via KIE.ai) — native joint video+audio
- Nano Banana 2 (Google, via KIE.ai) — face-locked scene editing
- Suno V3.5 (via KIE.ai) — instrumental underbeds
- ElevenLabs Speech-to-Speech (via direct API) — optional voice cloning
- FastAPI + Uvicorn — orchestration HTTP layer for n8n
- Pillow — slide typography rendering
- KIE.ai's `redpandaa.i.co` CDN — reliable asset hosting (replaces tmpfiles.org)

— Last updated 2026-05-14, Phase 3 daily-pack autopilot complete.

Challenges in Video Generation for the Persona

Suggested location in the capstone report: a "Challenges" section, placed before the proposed-solution / future-direction chapter that introduces the chosen commercial video service. ~1,400 words. This section characterises the problem only — it does not describe how we resolved it.

1. Problem Statement

A central goal of this project was to produce talking-head video clips of the synthetic persona for short-form video platforms (TikTok, Instagram Reels, YouTube Shorts). Of the four generative modalities required — text, image, voice, and video — three reached an acceptable level of consistency on the available compute. The text component, fine-tuned from a pretrained Phi-3.5-mini-instruct backbone, produced stylistically stable output. The image component, fine-tuned with a 30-photo LoRA over Z-Image-Turbo, achieved a measured ArcFace identity similarity of 0.664, a +120% lift over the pre-LoRA baseline of 0.301. The voice component, based on F5-TTS zero-shot cloning, reproduced the target voice from a single 9.77-second reference clip across arbitrary scripts.

The fourth modality, **audio-driven talking-head video**, did not converge to the same level of consistency. This section documents the failure modes we observed, the hardware constraints that bounded the search, and the specific gaps that remain unresolved.

2. What "Inconsistent" Means in This Context

The video inconsistency we observed was not at the level of identity drift — the face LoRA, used as the source still for every video, preserved a single recognisable face across generations. The inconsistency manifested at three other levels.

Inconsistent motion quality across models. Different audio-driven talking-head models produce qualitatively different kinds of "motion". Wav2Lip (Prajwal et al., 2020) produces sharp lip movements but leaves the rest of the head and body essentially frozen, making the output read as a still photo with moving lips. SadTalker (Zhang et al., 2023) produces natural head turns, blinks, and micro-expressions but softens the lip region. Neither, alone, produces output that resembles contemporary commercial talking-head video products such as HeyGen, Synthesia, or D-ID.

Inconsistent render-time vs. quality tradeoffs. Identical configurations (model, source image, audio file) produced highly variable wall-clock render times depending on system memory pressure. We measured one continuous render's iteration pace degrade from 33 s/iter to 1,294 s/iter — a 39× slowdown — as macOS swap usage rose from negligible to 11.4 GB out of 13 GB available. This made it impossible to plan a content pipeline around predictable wall-clock budgets: a clip that took 7 minutes on a fresh boot could take over two hours mid-session.

Inconsistent perceived realism across clip lengths. A short clip (3–5 seconds) of a model's talking-head output reads as reasonably natural. A 60–70-second clip of the same generation method reveals the underlying repetition in head motion patterns and the model's inability to gesture, change scene, or vary background — breaking the illusion that the platform algorithms select against.

These three failure modes share a root cause: the underlying audio-driven video models were not built for, and do not run well on, the hardware we had available.

3. The Apple Silicon Compute Ceiling

This project was developed on a single M-series Apple Silicon laptop (unified memory, no NVIDIA GPU). For text, image, and voice generation this is a workable platform — Apple's MLX framework supports the foundation models we used, and inference times for those modalities were measured in seconds or minutes. For video generation the same hardware is a substantially harder constraint.

The 2024–25 generation of audio-driven talking-head and full-body animators — Hallo3 (Cui et al., 2025), EchoMimic v2 and v3 (Meng et al., 2024), MuseTalk 1.5 (Tencent, 2025), JoyVASA (JD AI, 2024), and LiveAvatar (Alibaba, 2025) — share a common architectural pattern: diffusion-based latent video generation conditioned on audio features and a face image. This architecture has two consequences for our setting.

Memory. A 1024×1024×24fps video diffusion forward pass at the resolutions these models target requires 40–80 GB of GPU VRAM. Apple Silicon's unified memory architecture nominally exposes more memory than this, but the Metal Performance Shaders (MPS) backend in PyTorch does not yet implement the full set of operations these models depend on, and several of the models contain custom CUDA kernels with no MPS equivalent. We confirmed this empirically: each candidate project's official repository documents NVIDIA-only support, and the community-hosted HuggingFace Spaces hosting them (e.g., [fffiloni/echomimic-v2](#)) are explicitly labelled "CPU showcase" and return an `AppError` when actual inference is requested.

Throughput. Even when an MPS port can be coerced into running, it does so without the tensor-core acceleration that NVIDIA H100s use for the dominant compute pattern in diffusion models. The published real-time-on-H100 performance of these systems degrades to multiple minutes per second of output on Apple Silicon when a port runs at all.

The audio-driven video models we *could* run on Apple Silicon were older — the 2020 and 2023-vintage models cited above. Their output quality is visibly below what a 2025 reviewer expects from the term "AI talking head", and using them in production was not viable for the project's stated goal of producing native-feeling content for commercial platforms.

4. The Speed–Quality Frontier We Mapped

To characterise the achievable quality on our hardware, we measured five distinct configurations of the available models against the same 3.68-second reference audio. The table below summarises the results.

Configuration	Render time (3.68 s audio)	Notes
Wav2Lip (Improved quality, 256×256)	27 s	Sharp lip sync, identity preserved, no head motion or expression
SadTalker default (256×256, motion-enabled, no enhancer)	≈ 7 min	Natural head turns and blinks, lip sync visibly soft
SadTalker max (512×512, GFPGAN enhancer, motion suppressed)	75 min	High face fidelity, no head motion
Stacked: SadTalker motion + Wav2Lip refinement	24 min	Head motion + sharp lips — best single-model result available
Stacked, scaled to a 43-second TikTok script (extrapolated)	≈ 4 hours	Quality identical, wall-clock time prohibitive for routine production

Two findings emerged.

No single configuration dominated. Wav2Lip was fast and sharp at the lips but flat from the neck up. SadTalker produced motion but smudged the mouth. The best available combination required composing the two and multiplied the render cost.

Render time scaled super-linearly with audio length on this hardware, because longer renders held memory pressure higher for longer, which triggered the swap-thrashing degradation described in §2. A 71-second YouTube Shorts script under the highest-quality SadTalker configuration would have taken closer to six hours than a linear extrapolation suggests.

The strongest result the project's compute envelope allowed was the 24-minute stacked render, and only at clip lengths under 15 seconds. Routine production of platform-native content (TikTok at 15–60 sec, Reels at 30–90 sec, Shorts at 30–60 sec) was not feasible at this quality level.

5. Open Challenges

Three problems remain unresolved at the time of writing.

Realism gap relative to commercial systems. None of the locally-runnable models produce output that a contemporary social-media viewer would mistake for genuine human-recorded video. The visible artefacts (soft lips, frozen torso, repetitive head motion, background inertia) read as "AI-generated" within the first second of viewing. This is the gap between 2020-vintage and 2025-vintage talking-head technology, and on the available hardware that gap is not closeable.

Compute-bound throughput. Even at the achievable quality level, the wall-clock cost of a single 60-second clip exceeds what an autonomous content pipeline can sustain. The persona was originally designed to publish multiple posts per week across six platforms; a single video post at the highest local quality would consume multiple hours of compute and lock the workstation against any other use.

Body and scene variation. Every locally-runnable model we tested operates on a fixed face crop. The persona never walks, gestures with their hands, sits at a desk, or appears in any scene other than a medium-shot from the neck up. This is structurally inadequate for short-form video formats, where pattern-interrupt scene changes are a genre convention.

These three challenges, taken together, define the boundary between what was achievable on this project's compute and what the target content quality requires. They motivated the Phase 2 architecture described below.

6. Phase 2 Resolution: Kling API + Local Post-Production

Phase 2 addressed all three open challenges by splitting the pipeline into a cloud video stage and a local post-production stage.

Cloud video generation (Kling V1-6 API). The Kling multi-image-to-video endpoint accepts 1-4 reference photographs of the persona and a rich scene prompt, producing a 5- or 10-second 9:16 video in which the persona appears in the described environment with natural body language, scene depth, and camera motion. The Kling lip-sync endpoint then synchronises the persona's mouth to a provided audio file. Both calls run on Kling's Singapore cluster and return results in 2-4 minutes regardless of local hardware.

This resolved the three open challenges directly:

Challenge	Phase 1 State	Phase 2 State
Realism gap	Wav2Lip/SadTalker artefacts visible within 1 second	Kling-generated motion, scene, and lip-sync; ArcFace identity 0.696 average across frames
Compute throughput	4+ hours for a 60 s clip on Apple Silicon	7 minutes end-to-end for a 10 s reel (including voice + captions + assembly)
Body and scene variation	Fixed face crop, no environment	Full-body medium shots in varied real-world environments (café, office, rooftop)

Local post-production pipeline. Three components remain fully local, preserving the project's on-device generation constraint for persona-locked assets:

1. **F5-TTS voice cloning** — generates Daniel's voice from text using the 9.77 s reference clip. Runs on Apple Silicon via the local backend (port 8765). Output uploaded to catbox.moe for Kling's lip-sync endpoint.
2. **Whisper caption generation** — word-level timestamps from the generated audio, formatted as ASS subtitles with CapCut-style pop animation and keyword highlighting.
3. **FFmpeg reel builder** — concatenates clips, burns ASS captions, adds a hook text overlay with fade animation, and mixes background music at -22 dB. Produces a publish-ready H.264 MP4.

Identity verification. ArcFace scores across frames of the Phase 2 reel ranged from 0.672 to 0.710 (mean 0.695), compared to the Phase 1 LoRA-generated still peak of 0.664. The Kling multi-image reference approach matched or exceeded the fine-tuned LoRA for identity consistency.

Cost. Kling API usage is approximately 0.8 units per second of video at standard tier. A 10-second base clip plus lip-sync costs roughly 12-16 units total. At the project's remaining balance of ~\$7.30, this supports approximately 15-20 additional production runs.

Remaining limitation. Kling's maximum clip duration is 10 seconds. Reels longer than 10 seconds require generating and concatenating multiple clips, doubling the API cost per additional segment. The reel builder's multi-clip concatenation supports this, but the per-segment identity consistency has not been validated at scale.

7. Phase 3 Resolution: Native Joint Generation + Daily Pack Autopilot

Phase 2 worked but kept three persistent issues. Phase 3 (May 14) addressed each with a different architectural fix.

Problem 1 — Lip-sync drift at the end of multi-segment reels. The Phase-2 pipeline generated a silent multi-image-to-video reel with Kling V1-6 then ran a *separate* lip-sync re-render against an ElevenLabs voice track. Because the lip-sync pass treated the input video as a starting point and re-rendered the face per frame against a new audio timeline, accumulated sub-frame timing error showed visibly as drifting mouth-shape vs. spoken word by the back half of the 30-second reel.

Fix. Replaced the two-step pipeline with Kling 3.0 multi-shot (`sound: true`). This single endpoint *jointly* generates video, native speech, and lip-sync in one render — there is no compositing seam to drift. For 30-second reels we chain two 15-second Kling renders; concatenation now uses FFmpeg `aresample=async=1` with constant-rate re-encode at 48 kHz / 30 fps so the audio is locked to video timestamps across the join. Lip-sync now holds end-to-end.

Problem 2 — Static-background "AI portrait" feel. The Avatar-Pro approach (image + audio → talking video) animated only the face; the surrounding scene was a still image. The result reads as an AI-generated render within the first two seconds. Top short-form creators in the same niche (e.g. @andyhan.ai) explicitly use Veo 3 / Kling 3.0 for the opposite reason: native joint generation produces a *world* (passers-by, traffic, leaves, ambient motion) rather than a frozen photo with a moving mouth.

Fix. Kling 3.0 multi-shot with `kling_elements` for face-locking generates the entire world from scratch per render, including environmental motion. Additionally, Claude visual prompts now *require* explicit camera motion (handheld drift / push-in / dolly / orbit) and 2-3 named moving background elements; prompts that read as "locked-off" or "static" are rejected at script-validation time.

Problem 3 — Single-format output. Phase 2 produced reels only. Real creator feeds run a daily mix: reels for reach, portrait quotes for saves, carousels for swipe-through depth. A reel-only pipeline cannot sustain a creator account.

Fix. A new orchestrator, `scripts/daily_pack.py`, runs three content types in parallel:

- **Reel** — Claude 6-beat script → 2× Kling 3.0 multi-shot → audio resync → Suno underbed.
- **Portrait** — Claude headline + caption → Nano Banana 2 face-locked scene edit → Pillow typography overlay → 1080×1350 PNG.
- **Carousel** — Claude 5-slide story (cover / problem / insight / proof / payoff) → 5× Nano Banana → 5× Pillow slides.

All three pieces use distinct outfit / mood / setting / topic seeds within a single pack, with rotation tracked in `autopilot_history.json`. Wall-clock: ~12 min per pack; spend: ~\$5–7.

Reliability improvements that landed in Phase 3:

Issue	Fix
<code>tmpfiles.org</code> rate-limiting mid-job (Kling could not fetch supplied URLs)	Switched every KIE-bound upload to KIE's own <code>file-stream-upload</code> on the <code>redpandaai.co</code> CDN
Concept-slide static backgrounds	Visual-prompt validator rejects prompts without motion + named moving elements
Outfit / posture repetition across packs	<code>autopilot.py</code> rotation pools (8 outfits / 8 moods / 5 setting pools); each pack picks 3 distinct combos avoiding the last 4-6
Topics clustering on the same anecdote ("fired my X for AI")	14-entry topic seed list, with intra-pack avoid-list passed to each piece's Claude call

Phase 3 results. Two daily packs (six published pieces total) shipped to YouTube + Instagram + LinkedIn between 19:43 and 21:08 on May 14, with no human intervention. Lip-sync drift no longer observed. Background motion verified on both reels. All six pieces in the two packs use distinct outfits, moods, and topic seeds. The full pipeline is now exposed as HTTP endpoints (`scripts/api_server.py`) for n8n / Zapier / cron scheduling.

What Phase 3 leaves open. True 5-slide Instagram carousel (vs. single-image cover) requires bypassing Composio's `INSTAGRAM_CREATE_MEDIA_CONTAINER` validator and calling Meta's Graph API directly. X / Twitter publishing is blocked on a one-time Composio OAuth dance for the X toolkit. Optional ElevenLabs Speech-to-Speech voice-cloning to a specific target voice is wired in `produce_reel_v16.py` but requires an `ELEVENLABS_API_KEY` to activate; without it, packs ship with Kling 3.0's native voice.

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